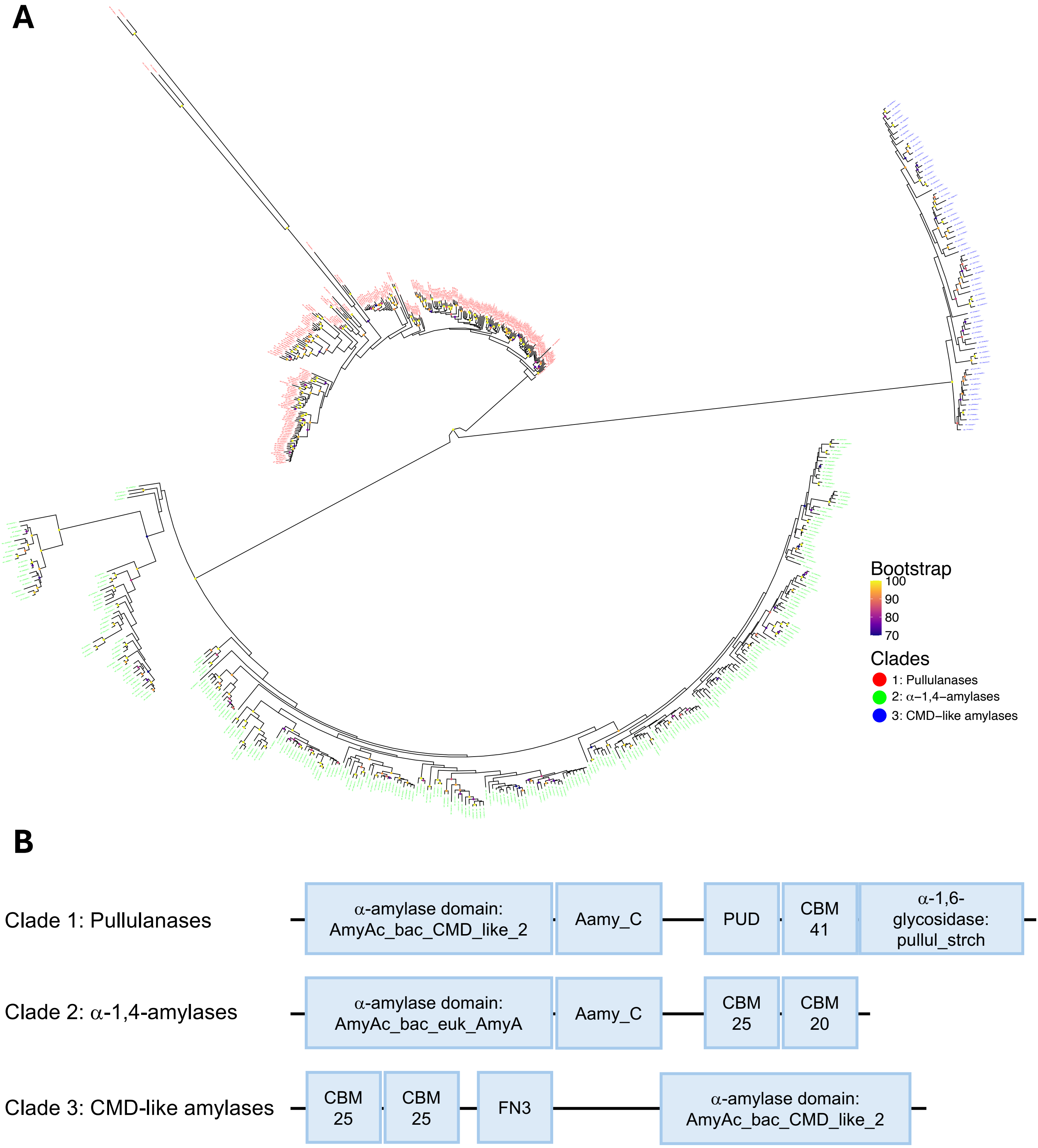


Figure 1: Streptomycetes possess three types of amylases. A total of 645 amylases were found across 295 *Streptomyces* genomes. Phylogenetic analysis showed that amylases in streptomycetes cluster into three major clades. Branch support values (Bootstrap) are shown as Felsenstein's Bootstrap Proportions scores, with coloured circles indicating values ≥ 70 % at internal nodes (A). Conserved domain analysis of representative proteins from each clade found that clade 1 consisted of pullulanases, clade 2 of α-1,4-acting amylases, and clade 3 of cyclomaltodextrin (CMD)-like amylases (B). AmyAc_bac_CMD_like_2: α-amylase catalytic domain found in bacterial cyclomaltodextrinases; Aamy_C: maltogenic C-terminal amylase domain; PUD: pullulanase associated domain; CBM: carbohydrate binding module; pullul_strch: pullulanase type α-1,6-glycosidase domain; AmyAc_bac_euk_AmyA: α-1,4-amylase catalytic domain found in bacterial and eukaryotic α-amylases; FN3: Fibronectin type 3 domain. Domains not drawn to scale.

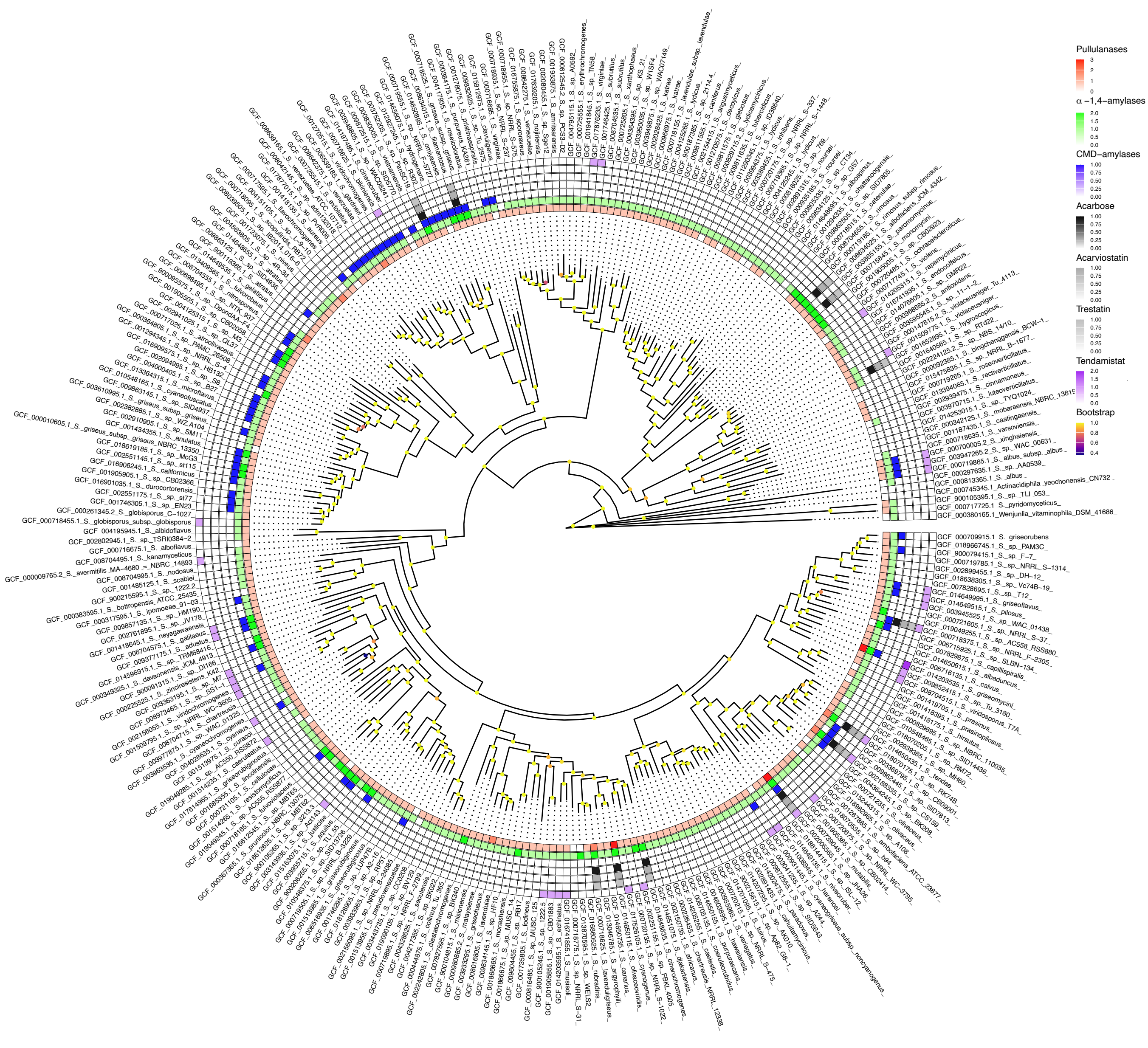


Figure 2: Amylases frequently co-occur with amylase inhibitors in *Streptomyces* genomes. The evolutionary relationship between 295 *Streptomyces* species were inferred from whole genome phylogeny and the prevalence of amylases and amylase inhibitors is visualised using a colour scale (white indicates absence; darker colours indicate higher prevalence). Annotations shown in the legend (right panel), listed from top to bottom, correspond to the concentric circles around the phylogenetic tree from the innermost to the outermost. Pullulanases (innermost, red) and α-1,4-amylases (green) are found almost ubiquitously throughout the *Streptomyces* genus, whereas only a subset of species encode cyclomaltodextrin-like amylases (blue). Genomes with higher numbers of amylases frequently coincide with those encoding amylase inhibitors, including acarbose (black), acarviostatin (darkgrey), trestatin (grey), and tendamistat (purple). Branch support values (Bootstrap), computed as Transfer Bootstrap Expectation scores, are represented by coloured circles at internal nodes.

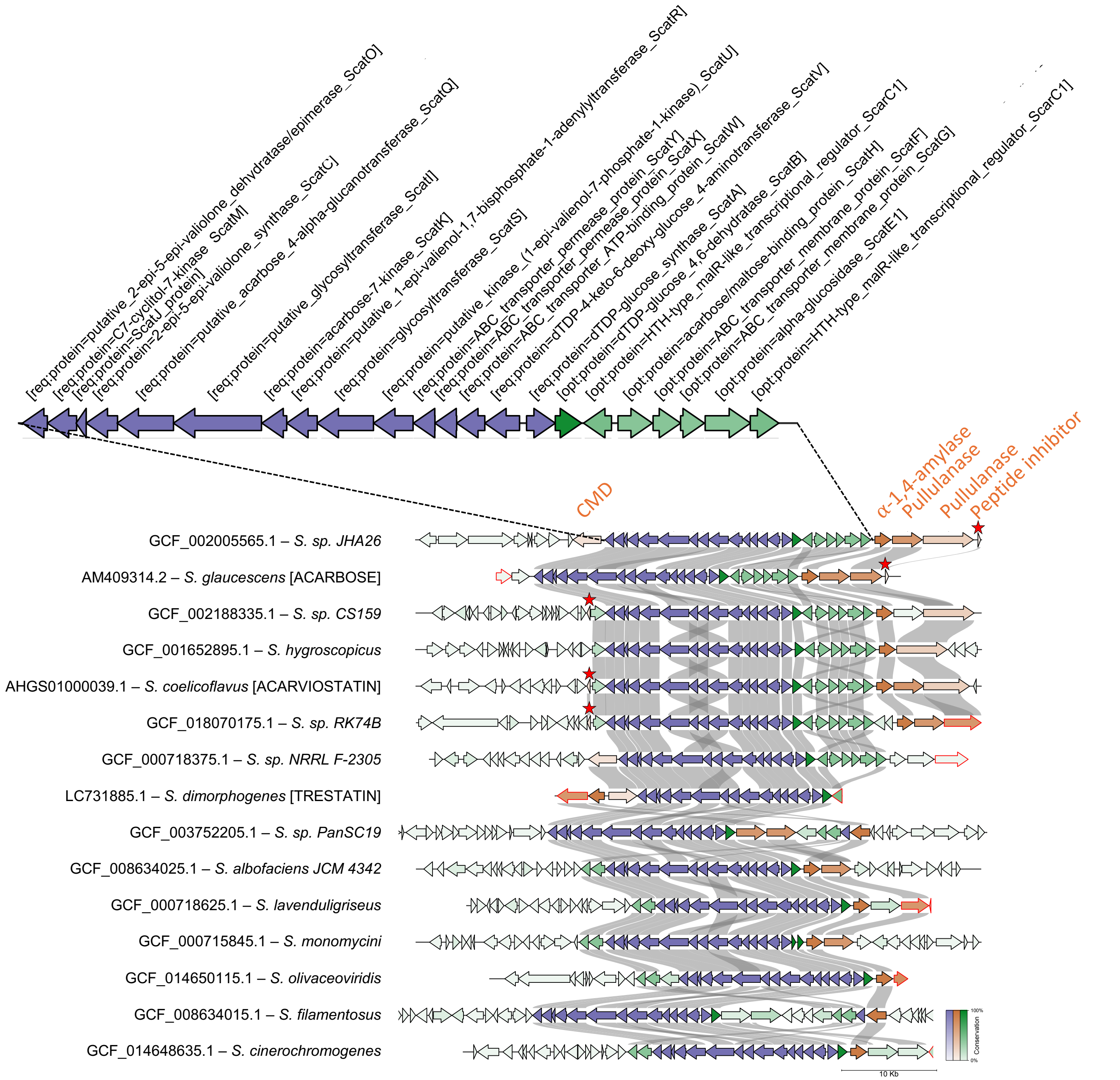


Figure 3: Acarbose-like BGCs possess between one and four amylases in their gene-neighbourhoods. GATOR identified 15 genomes (12 from the representative list of 295 genomes plus the acarviostatin, acarbose, and trestatin BGCs already described in literature). Core genes required for the synthesis of aminocyclitol are shown in blue, amylases are shown in orange, and other genes in green. Tendamistat-like peptide inhibitors are shown in orange and marked with a red star. Gene level conservation between clusters are shown by colour intensity (more intense colour for higher level of conservation). Genes with red edges indicate contig edges. CMD: cyclomaltodextrinase.

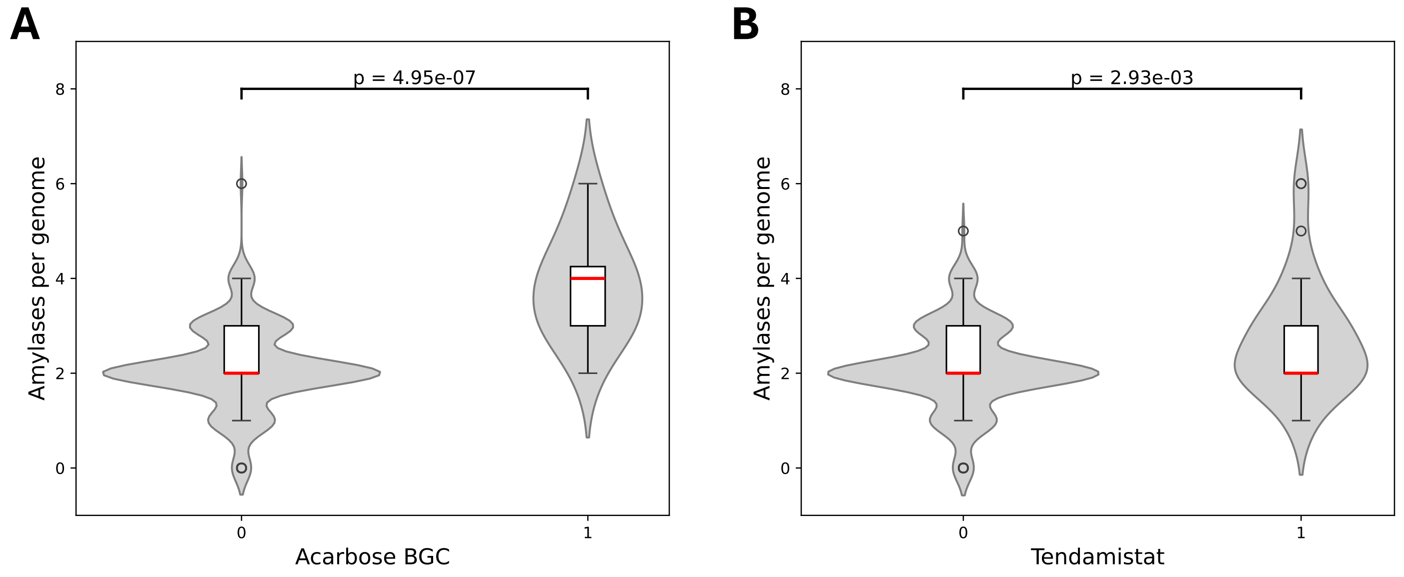


Figure 4: *Streptomyces* species encoding amylase inhibitors possess more amylase genes. Violin-boxplots showing significant correlations between higher numbers of amylase per genome and genomes encoding acarbose BGCs (A) [$r = 0.78$; $p = 4.95 \times 10^{-7}$] or tendamistat (B) [$r = 0.27$; $p = 2.93 \times 10^{-3}$].

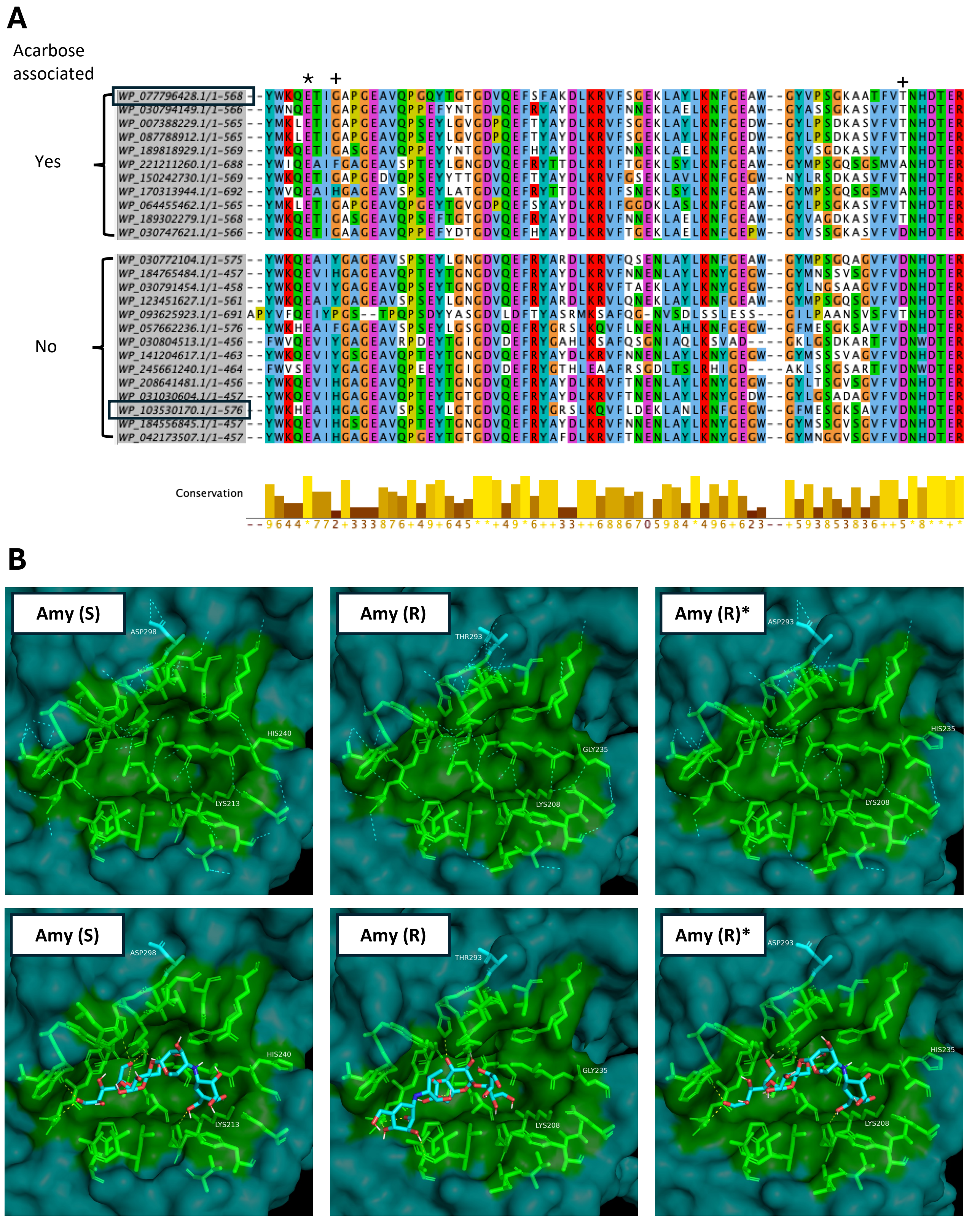


Figure 5: Predicted binding sites of acarbose differs between amylases that are located in acarbose BGCs and amylases that are not. MSA shows an enrichment of GLY235 and THR293 (+) in alpha-1,4-amylases located in acarbose BGCs. Both residues are found within the conserved catalytic pocket close to the catalytic residue glutamate (*) (A). The catalytic pockets (residues highlighted in green) are shown for a putatively acarbose-sensitive amylase [Amy (S) – accession: WP_103530170] found in a genome without acarbose and a putatively resistant amylase [Amy (R) – accession: WP_077796428] found within an acarbose BGC [proteins marked by black squares in (A)]. Amy (R)* shows the structure of Amy (R) where GLY235 and THR293 have been converted to HIS and ASP, identical to the residues found at these positions in Amy (S). Hydrogen bonds between residues within the catalytic pocket are shown with dashed cyan lines and hydrogen bonds between acarbose and the proteins are shown with dashed yellow lines (bottom panel). LYS213/208, previously described as involved in inferring acarbose resistance is highlighted (B).

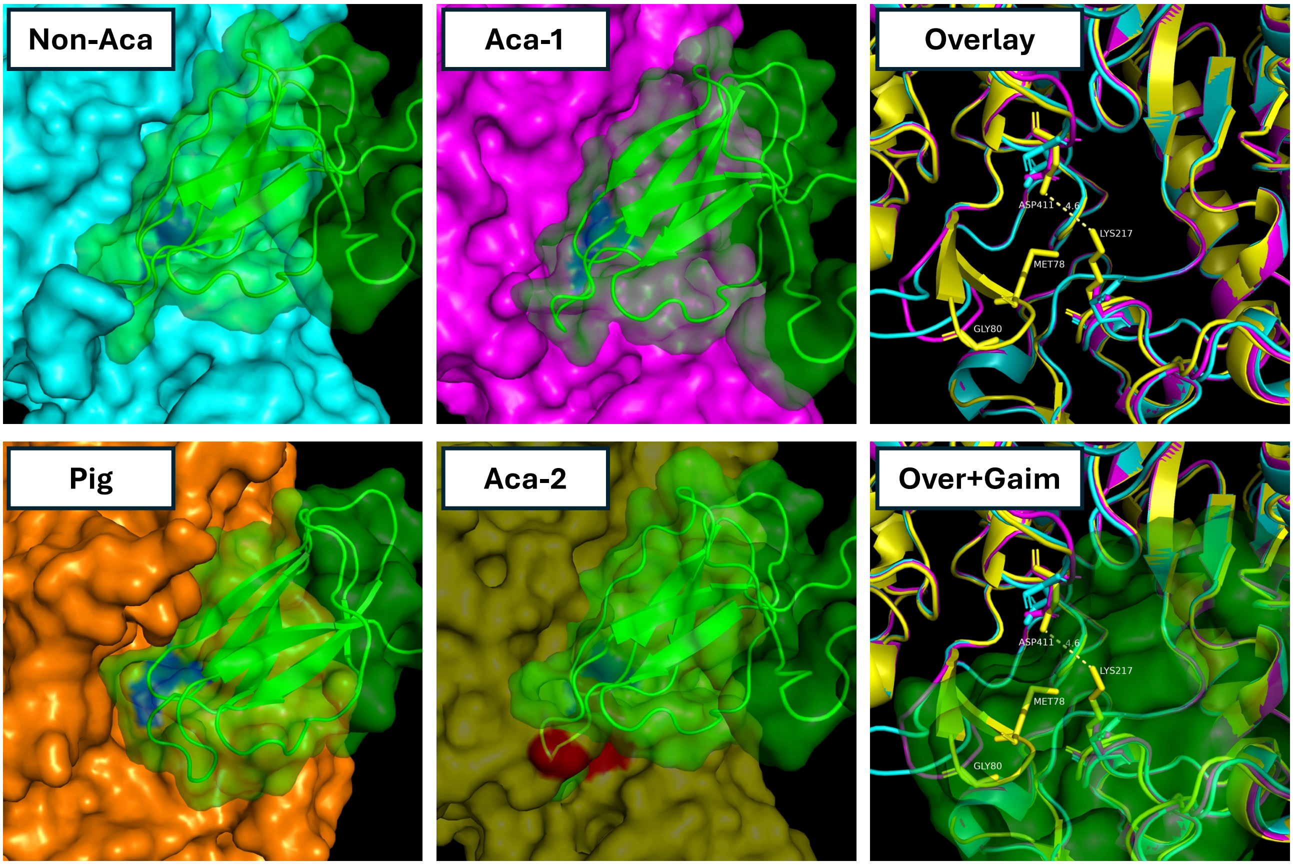
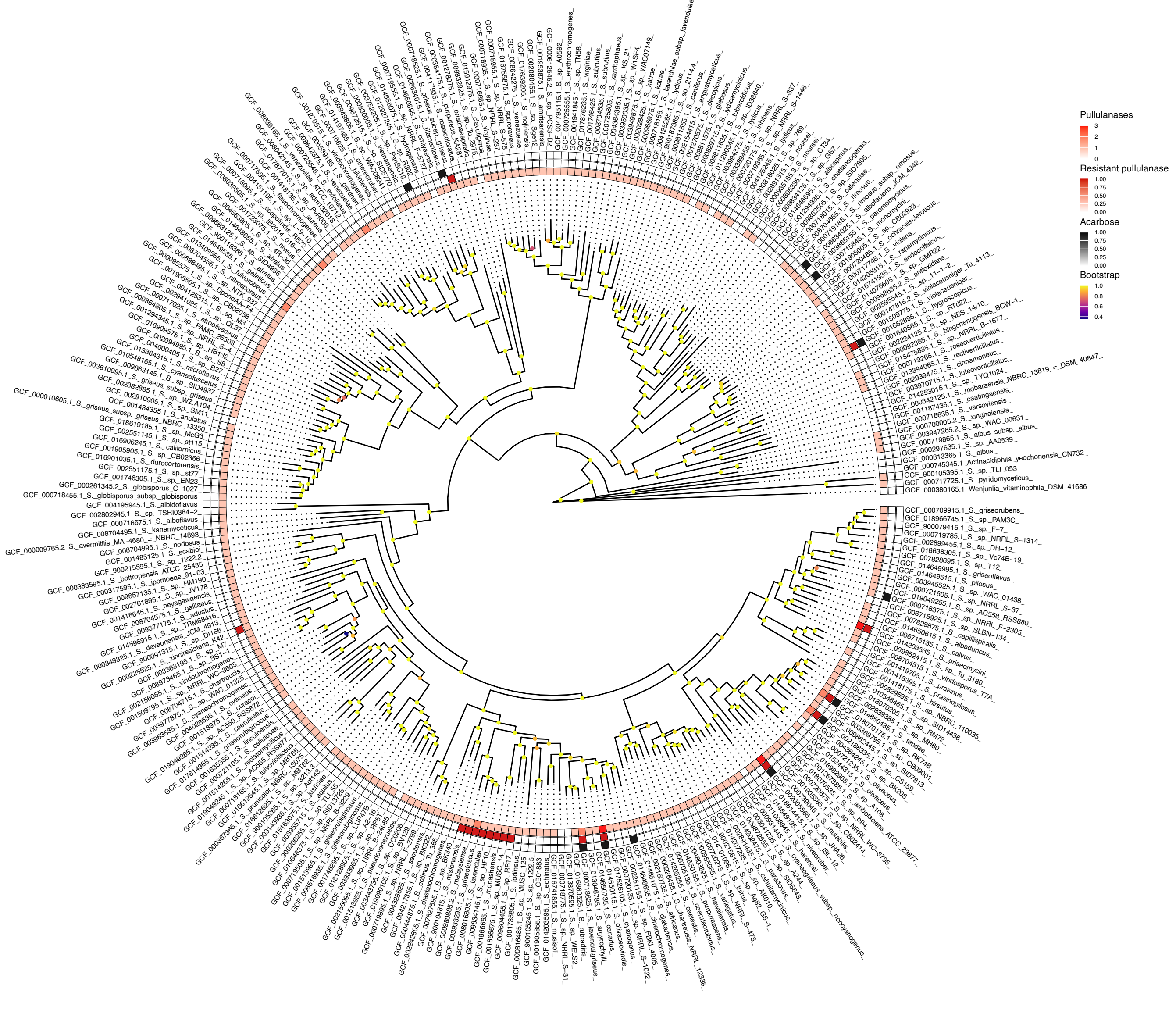
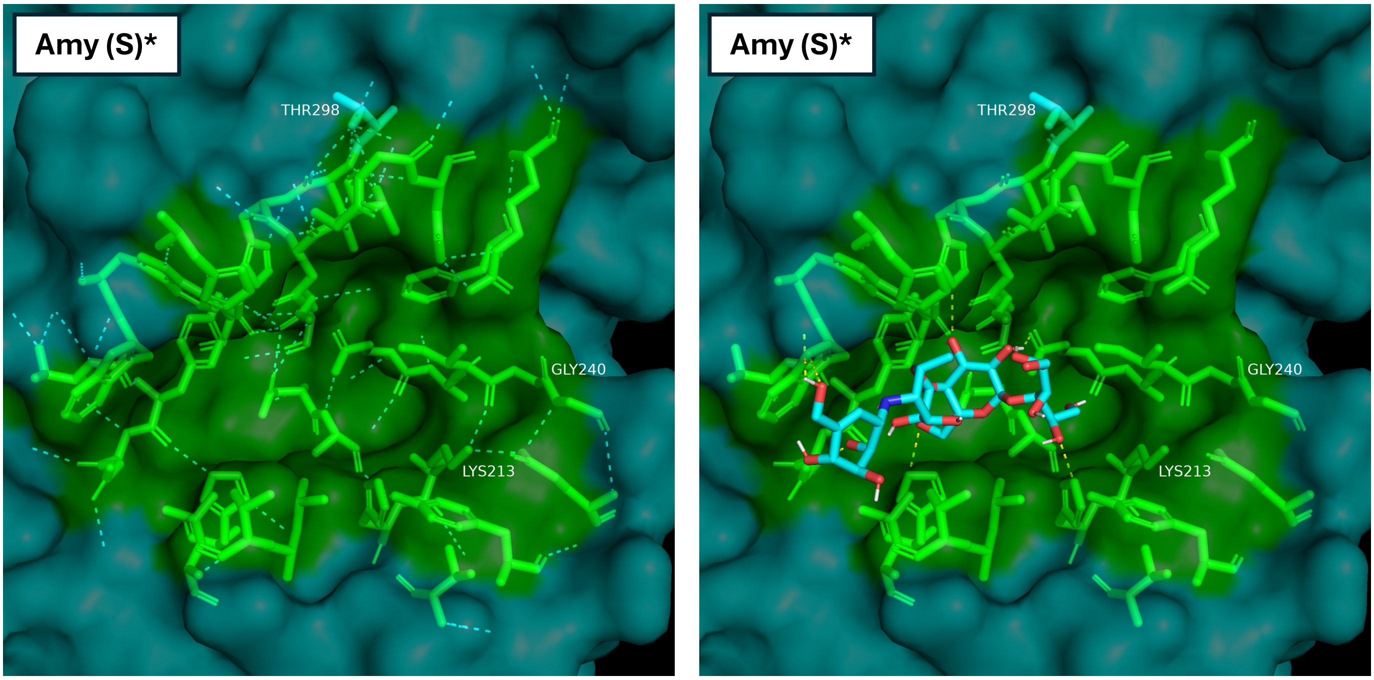


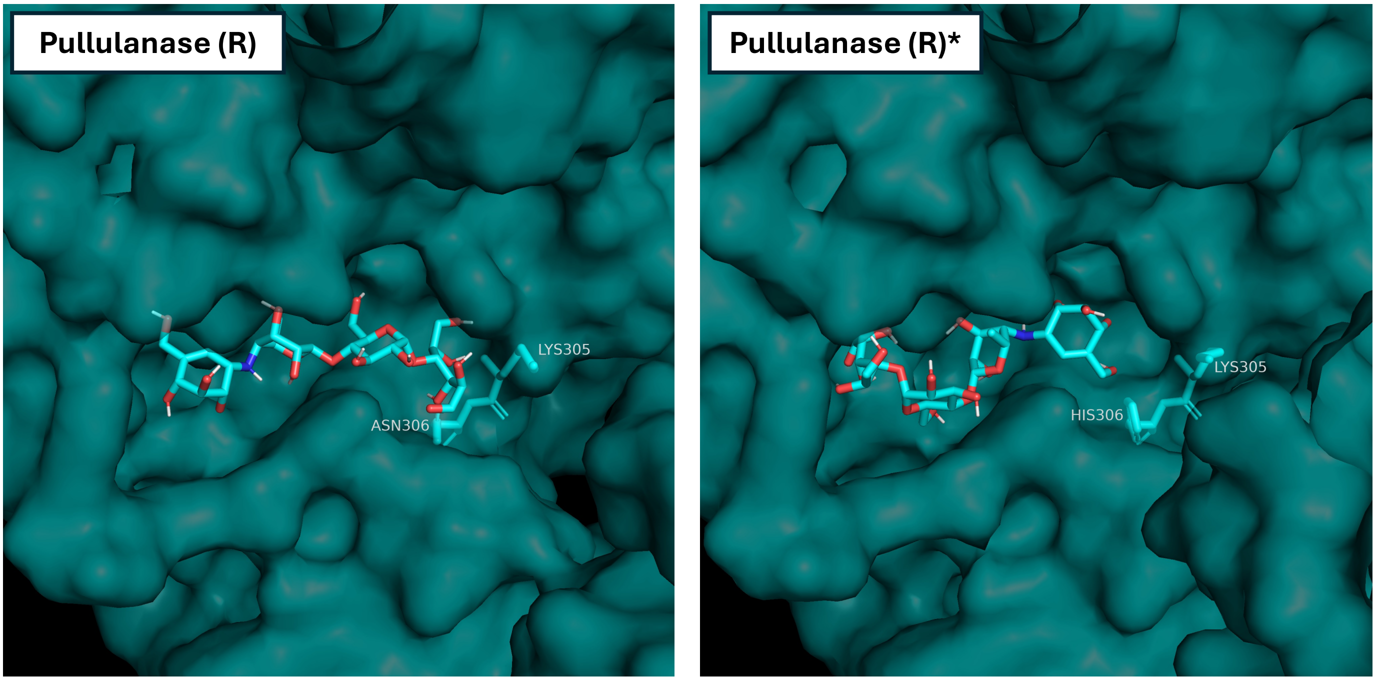
Figure 6: Structure predictions and protein-protein docking analyses suggest obstructed binding of proteinaceous amylase inhibitor Gaim to an acarbose-associated pullulanase from *Streptomyces sp. JHA26.* Gaim is predicted to bind two pullulanases from *Streptomyces sp. JHA26:* one encoded within the acarbose BGC (Aca-1 [magenta]; WP_077796430.1) and one located outside the BGC (Non-Aca [cyan]; WP_077796080.1). Gaim (green) encoded within the acarbose BGC is predicted to bind in the starch binding pocket containing the catalytic aspartate and glutamate residues (blue), in a manner similar to the binding of tendamistat (green) to porcine pancreatic amylase (orange; Pig). Alignment of the second acarbose BGC-encoded pullulanase (Aca-2 [transparent yellow]; WP_077796429.1) with the Gaim-bound non-acarbose-associated pullulanase reveals three protruding amino acid residues (red) that are predicted to obstruct Gaim binding. An overlay (Overlay) of the three *Streptomyces* amylases highlights these residues (MET78, GLY80, and LYS217), including weak electrostatic interactions between ASP411 and LYS217 which creates a closed pocket confirmation that sterically clashes with Gaim binding (Over+Gaim).



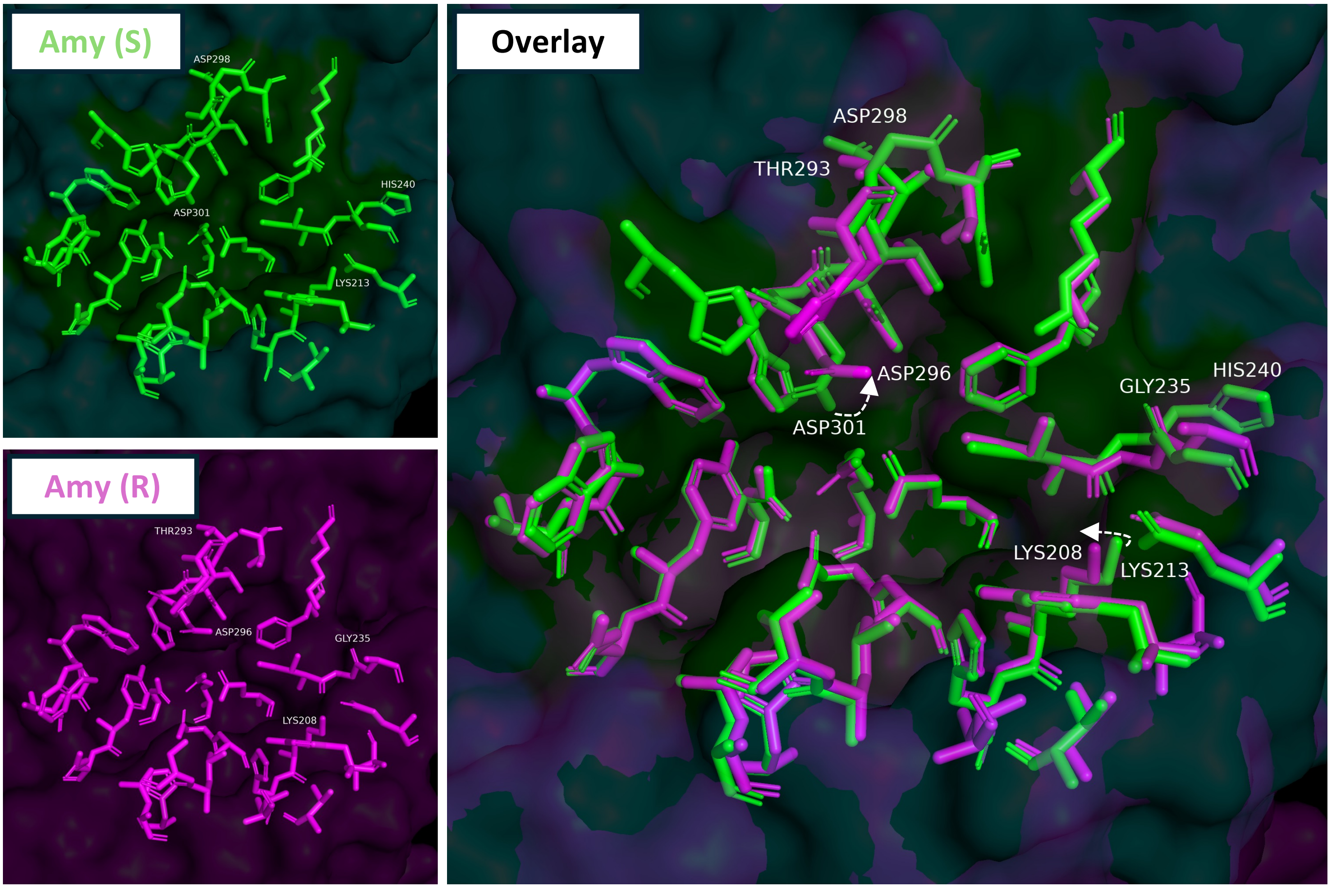
Supplementary Figure S1: Amylase inhibitor resistant amylases are enriched in genomes encoding acarbose BGCs.



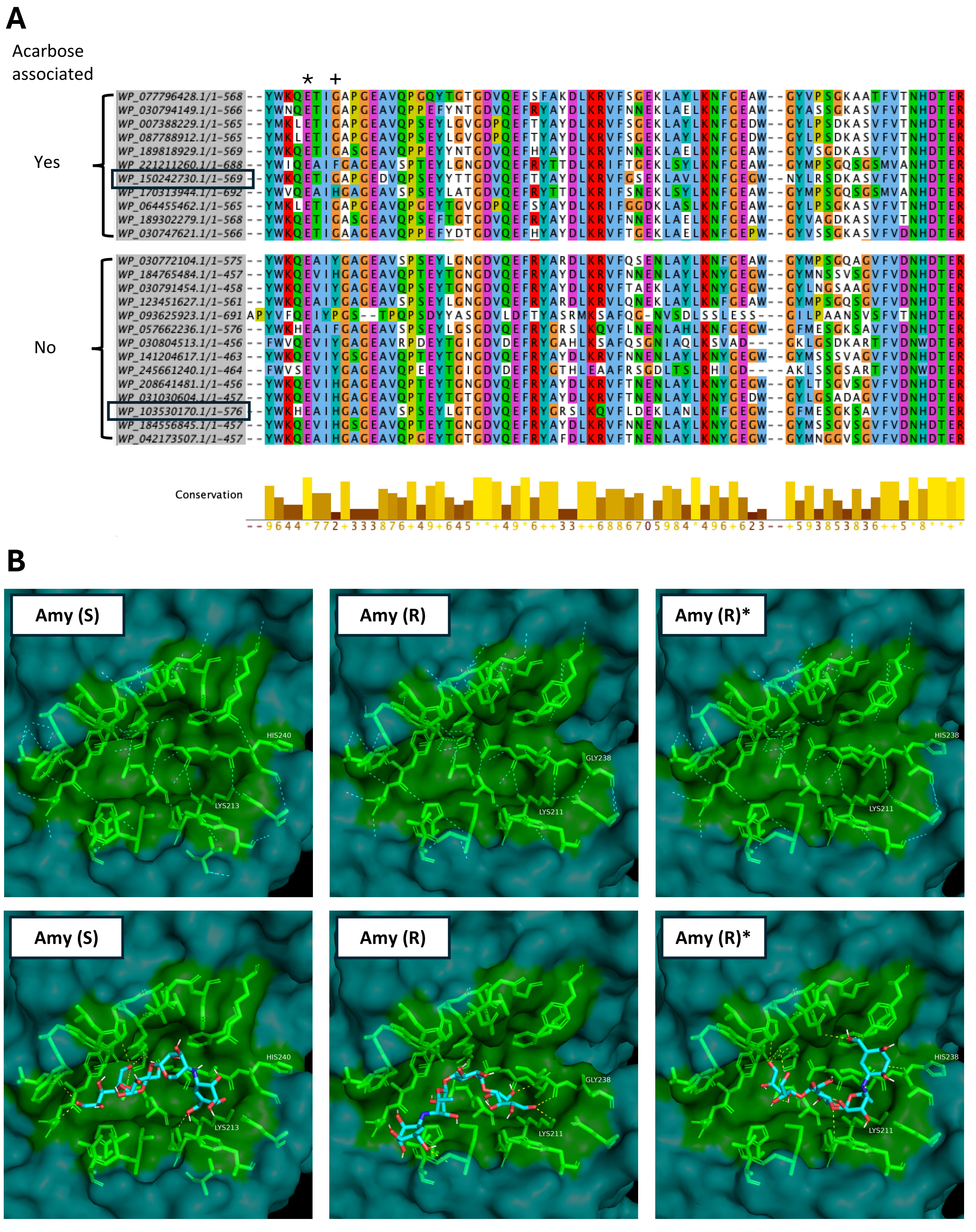
Supplementary Figure S2: D298T and H240G changes the binding pattern of acarbose in a non-acarbose-associated amylase (WP_103530170) to that seen in acarbose-associated amylases.



Supplementary Figure S3: N306H reorients the predicted binding of acarbose in an inhibitor resistant pullulanase (WP_064455463) to a configuration similar to that seen in alpha-1,4-amylases found in genomes without acarbose.



Supplementary Figure S4: The architecture of the catalytic pocket differs between Amy (S) and Amy (R). Amy (S) [WP_103530170; green] and Amy (R) [WP_077796428; magenta] varies on residue 293/298 and 235/240 which might be causative of the reorientation of D296/301 and K208/213 resulting in the narrower binding pocket found in Amy (R).



Supplementary Figure S5: G238H reorients acarbose binding in an α-1,4-amylase from an acarbose BGC without an acarbose resistant pullulanase. MSA shows an enrichment of GLY238 (+) in alpha-1,4-amylases located in acarbose BGCs (A). The catalytic pockets (residues highlighted in green) are shown for a putatively acarbose-sensitive amylase [Amy (S) – accession: WP_103530170] found in a genome without acarbose and a putatively resistant amylase [Amy (R) – accession: WP_150242730] found within an acarbose BGC [proteins marked by black squares in (A)]. Amy (R)* shows the structure of Amy (R) where GLY238 have been converted to HIS, identical to the residue found at this positions in Amy (S). Hydrogen bonds between residues within the catalytic pocket are shown with dashed cyan lines and hydrogen bonds between acarbose and the proteins are shown with dashed yellow lines (bottom panel). LYS213/211, previously described as involved in inferring acarbose resistance is highlighted (B).